Nordic Laboratories

PATIENT: **XXXXXXXXXXXXXXXX**			TEST REF: **TST-NL-XXXX**
TEST NUMBER: T-NL-XXXXX (XXXXXXXXXX)		COLLECTED: XX/XX/XXXX	PRACTITIONER: **XXXXXXXXXXXX**
GENDER: XYZ		RECEIVED: XX/XX/XXXX	
AGE: XX		TESTED: XX/XX/XXXX	XXXXXXXXXXXXXXXXXXXX

TEST NAME: Metabolomix+

3200 Metabolomix+ - FMV Urine

Results Overview

Functional Imbalance Scores

Key: 0-4 : Minimal Need for Support; 5-7 : Moderate Need for Support; 8-10 : High Need for Support

Need for Antioxidant Support	Need for Mitochondrial Support	Need for Reduced Exposure	Need for Methylation Support
Oxidative Stress	Mitochondrial Dysfunction	Toxic Exposure	Methylation Imbalance
4	7	8	2
Cystine ●	FIGLU ●	α-Hydroxyisobutyric Acid ▲	Methylmalonic Acid ●
Cysteine ●	Methylmalonic Acid ●	α-Ketophenylacetic Acid △	Methionine ●
Lipid Peroxides ●	Glutaric Acid △	Pyroglutamic Acid ▲	FIGLU ●
8-OHdG ●	Lactic Acid ●	Orotic Acid ●	Sarcosine ●
Taurine △	Pyruvic Acid ▽	Citric Acid ▲	Vanilmandelic Acid ●
Citric Acid ▲	Citric Acid ▲	cis-Aconitic Acid ●	Arginine ▼
cis-Aconitic Acid ●	cis-Aconitic Acid ●	Isocitric Acid ▲	Glycine ●
	Isocitric Acid ▲	Glutaric Acid △	Serine ●
	α-Ketoglutaric Acid △		Creatinine ●
	Succinic Acid △		
	Malic Acid ▲		
	Adipic Acid ▲		
	Suberic Acid △		

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TEST NAME: Metabolomix+

Nutrient Need Overview

	Nutrient Need (0 1 2 3 4 5 6 7 8 9 10)	DRI	Suggested Recommendations	Provider Recommendations
Antioxidants				
Vitamin A		2,333 IU	3,000 IU	
Vitamin C		75 mg	250 mg	
Vitamin E / Tocopherols		22 IU	100 IU	
α-Lipoic Acid			100 mg	
CoQ10			60 mg	
Glutathione				
Plant-based Antioxidants				
B-Vitamins				
Thiamin - B1		1.1 mg	25 mg	
Riboflavin - B2		1.1 mg	50 mg	
Niacin - B3		14 mg	50 mg	
Pyridoxine - B6		1.5 mg	25 mg	
Biotin - B7		30 mcg	400 mcg	
Folic Acid - B9		400 mcg	400 mcg	
Cobalamin - B12		2.4 mcg	100 mcg	
Minerals				
Magnesium		320 mg	800 mg	
Manganese		1.8 mg	3.0 mg	
Molybdenum		45 mcg	75 mcg	
Zinc		8 mg	10 mg	
GI Support				
Digestive Support/Enzymes			0 IU	
Microbiome Support/Probiotics			10 billion CFU	

Amino Acids (mg/day)

Arginine	1,314	Methionine	0
Asparagine	0	Phenylalanine	0
Cysteine	0	Serine	0
Glutamine	89	Taurine	0
Glycine	0	Threonine	0
Histidine	0	Tryptophan	0
Isoleucine	0	Tyrosine	0
Leucine	0	Valine	0
Lysine	74		

Recommendations for age and gender-specific supplementation are set by comparing levels of nutrient functional need to optimal levels as described in the peer-reviewed literature. They are provided as guidance for short-term support of nutritional deficiencies only.

The Nutrient Need Overview is provided at the request of the ordering practitioner. Any application of it as a therapeutic intervention is to be determined by the ordering practitioner.

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TEST NAME: Metabolomix+

Interpretation At-A-Glance

Antioxidant Needs

Vitamin A

- Beta-carotene & other carotenoids are converted to vitamin A (retinol), involved in vision, antioxidant & immune function, gene expression & cell growth.
- Vitamin A deficiency may occur with chronic alcoholism, zinc deficiency, hypothyroidism, or oral contraceptives containing estrogen & progestin.
- Deficiency may result in night blindness, impaired immunity, healing & tissue regeneration, increased risk of infection, leukoplakia or keratosis.
- Food sources include cod liver oil, fortified cereals & milk, eggs, sweet potato, pumpkin, carrot, cantaloupe, mango, spinach, broccoli, kale & butternut squash.

Vitamin C

- Vitamin C is an antioxidant (also used in the regeneration of other antioxidants). It is involved in cholesterol metabolism, the production & function of WBCs and antibodies, and the synthesis of collagen, norepinephrine and carnitine.
- Deficiency may occur with oral contraceptives, aspirin, diuretics or NSAIDs.
- Deficiency can result in scurvy, swollen gingiva, periodontal destruction, loose teeth, sore mouth, soft tissue ulcerations, or increased risk of infection.
- Food sources include oranges, grapefruit, strawberries, tomato, sweet red pepper, broccoli and potato.

Vitamin E / Tocopherols

- Alpha-tocopherol (body's main form of vitamin E) functions as an antioxidant, regulates cell signaling, influences immune function and inhibits coagulation.
- Deficiency may occur with malabsorption, cholestyramine, colestipol, isoniazid, orlistat, olestra and certain anti-convulsants (e.g., phenobarbital, phenytoin).
- Deficiency may result in peripheral neuropathy, ataxia, muscle weakness, retinopathy, and increased risk of CVD, prostate cancer and cataracts.
- Food sources include oils (olive, soy, corn, canola, safflower, sunflower), eggs, nuts, seeds, spinach, carrots, avocado, dark leafy greens and wheat germ.

α-Lipoic Acid

- α-Lipoic acid plays an important role in energy production, antioxidant activity (including the regeneration of vitamin C and glutathione), insulin signaling, cell signaling and the catabolism of α-keto acids and amino acids.
- High biotin intake can compete with lipoic acid for cell membrane entry.
- Optimal levels of α-lipoic acid may improve glucose utilization and protect against diabetic neuropathy, vascular disease and age-related cognitive decline.
- Main food sources include organ meats, spinach and broccoli. Lesser sources include tomato, peas, Brussels sprouts and brewer's yeast.

CoQ10

- CoQ10 is a powerful antioxidant that is synthesized in the body and contained in cell membranes. CoQ10 is also essential for energy production & pH regulation.
- CoQ10 deficiency may occur with HMG-CoA reductase inhibitors (statins), several anti-diabetic medication classes (biguanides, sulfonylureas) or beta-blockers.
- Low levels may aggravate oxidative stress, diabetes, cancer, congestive heart failure, cardiac arrhythmias, gingivitis and neurologic diseases.
- Main food sources include meat, poultry, fish, soybean, canola oil, nuts and whole grains. Moderate sources include fruits, vegetables, eggs and dairy.

Glutathione

- Glutathione (GSH) is composed of cysteine, glutamine & glycine. GSH is a source of sulfate and plays a key role in antioxidant activity and detoxification of toxins.
- GSH requirement is increased with high-fat diets, cigarette smoke, cystinuria, chronic alcoholism, chronic acetaminophen use, infection, inflammation and toxic exposure.
- Deficiency may result in oxidative stress & damage, impaired detoxification, altered immunity, macular degeneration and increased risk of chronic illness.
- Food sources of GSH precursors include meats, poultry, fish, soy, corn, nuts, seeds, wheat germ, milk and cheese.

Plant-based Antioxidants

- Oxidative stress is the imbalance between the production of free radicals and the body's ability to readily detoxify these reactive species and/or repair the resulting damage with anti-oxidants.
- Oxidative stress can be endogenous (energy production and inflammation) or exogenous (exercise, exposure to environmental toxins).
- Oxidative stress has been implicated clinically in the development of neurodegenerative diseases, cardiovascular diseases and chronic fatigue syndrome.
- Antioxidants may be found in whole food sources (e.g., brightly colored fruits & vegetables, green tea, turmeric) as well as nutraceuticals (e.g., resveratrol, EGCG, lutein, lycopene, ginkgo, milk thistle, etc.).

KEY

Function of Nutrient | Cause of Deficiency | Complications of Deficiency | Food Sources of Nutrient

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TEST NAME: Metabolomix+

Interpretation At-A-Glance

B-Vitamin Needs

Thiamin - B1

- B1 is a required cofactor for enzymes involved in energy production from food, and for the synthesis of ATP, GTP, DNA, RNA and NADPH.
- Low B1 can result from chronic alcoholism, diuretics, digoxin, oral contraceptives and HRT, or large amounts of tea & coffee (contain anti-B1 factors).
- B1 deficiency may lead to dry beriberi (e.g., neuropathy, muscle weakness), wet beriberi (e.g., cardiac problems, edema), encephalopathy or dementia.
- Food sources include lentils, whole grains, wheat germ, Brazil nuts, peas, organ meats, brewer's yeast, blackstrap molasses, spinach, milk & eggs.

Pyridoxine - B6

- B6 (as P5P) is a cofactor for enzymes involved in glycogenolysis & gluconeogenesis, and synthesis of neurotransmitters, heme, B3, RBCs and nucleic acids.
- Low B6 may result from chronic alcoholism, long-term diuretics, estrogens (oral contraceptives and HRT), anti-TB meds, penicillamine, L-DOPA or digoxin.
- B6 deficiency may result in neurologic symptoms (e.g., irritability, depression, seizures), oral inflammation, impaired immunity or increased homocysteine.
- Food sources include poultry, beef, beef liver, fish, whole grains, wheat germ, soybean, lentils, nuts & seeds, potato, spinach and carrots.

Riboflavin - B2

- B2 is a key component of enzymes involved in antioxidant function, energy production, detoxification, methionine metabolism and vitamin activation.
- Low B2 may result from chronic alcoholism, some anti-psychotic medications, oral contraceptives, tricyclic antidepressants, quinacrine or adriamycin.
- B2 deficiency may result in oxidative stress, mitochondrial dysfunction, low uric acid, low B3 or B6, high homocysteine, anemia or oral & throat inflammation.
- Food sources include milk, cheese, eggs, whole grains, beef, chicken, wheat germ, fish, broccoli, asparagus, spinach, mushrooms and almonds.

Biotin - B7

- Biotin is a cofactor for enzymes involved in functions such as fatty acid synthesis, mitochondrial FA oxidation, gluconeogenesis and DNA replication & transcription.
- Deficiency may result from certain inborn errors, chronic intake of raw egg whites, long-term TPN, anticonvulsants, high-dose B5, sulfa drugs & other antibiotics.
- Low levels may result in neurologic symptoms (e.g., paresthesias, depression), hair loss, scaly rash on face or genitals or impaired immunity.
- Food sources include yeast, whole grains, wheat germ, eggs, cheese, liver, meats, fish, wheat, nuts & seeds, avocado, raspberries, sweet potato and cauliflower.

Niacin - B3

- B3 is used to form NAD and NADP, involved in energy production from food, fatty acid & cholesterol synthesis, cell signaling, DNA repair & cell differentiation.
- Low B3 may result from deficiencies of tryptophan (B3 precursor), B6, B2 or Fe (cofactors in B3 production), or from long-term isoniazid or oral contraceptive use.
- B3 deficiency may result in pellagra (dermatitis, diarrhea, dementia), neurologic symptoms (e.g., depression, memory loss), bright red tongue or fatigue.
- Food sources include poultry, beef, organ meats, fish, whole grains, peanuts, seeds, lentils, brewer's yeast and lima beans.

Folic Acid - B9

- Folic acid plays a key role in coenzymes involved in DNA and SAMe synthesis, methylation, nucleic acids & amino acid metabolism and RBC production.
- Low folate may result from alcoholism, high-dose NSAIDs, diabetic meds, H2 blockers, some diuretics and anti-convulsants, SSRIs, methotrexate, trimethoprim, pyrimethamine, triamterene, sulfasalazine or cholestyramine.
- Folate deficiency can result in anemia, fatigue, low methionine, increased homocysteine, impaired immunity, heart disease, birth defects and CA risk.
- Food sources include fortified grains, green vegetables, beans & legumes.

Cobalamin - B12

- B12 plays important roles in energy production from fats & proteins, methylation, synthesis of hemoglobin & RBCs, and maintenance of nerve cells, DNA & RNA.
- Low B12 may result from alcoholism, malabsorption, hypochlorhydria (e.g., from atrophic gastritis, H. pylori infection, pernicious anemia, H2 blockers, PPIs), vegan diets, diabetic meds, cholestyramine, chloramphenicol, neomycin or colchicine.
- B12 deficiency can lead to anemia, fatigue, neurologic symptoms (e.g., paresthesias, memory loss, depression, dementia), methylation defects or chromosome breaks.
- Food sources include shellfish, red meat poultry, fish, eggs, milk and cheese.

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TEST NAME: Metabolomix+

Interpretation At-A-Glance

Mineral Needs

Magnesium

- Magnesium is involved in >300 metabolic reactions. Key areas include energy production, bone & ATP formation, muscle & nerve conduction and cell signaling.
- Deficiency may occur with malabsorption, alcoholism, hyperparathyroidism, renal disorders (wasting), diabetes, diuretics, digoxin or high doses of zinc.
- Low Mg may result in muscle weakness/spasm, constipation, depression, hypertension, arrhythmias, hypocalcemia, hypokalemia or personality changes.
- Food sources include dark leafy greens, oatmeal, buckwheat, unpolished grains, chocolate, milk, nuts & seeds, lima beans and molasses.

Manganese

- Manganese plays an important role in antioxidant function, gluconeogenesis, the urea cycle, cartilage & bone formation, energy production and digestion.
- Impaired absorption of Mn may occur with excess intake of Fe, Ca, Cu, folic acid, or phosphorous compounds, or use of long-term TPN, Mg-containing antacids or laxatives.
- Deficiency may result in impaired bone/connective tissue growth, glucose & lipid dysregulation, infertility, oxidative stress, inflammation or hyperammonemia.
- Food sources include whole grains, legumes, dried fruits, nuts, dark green leafy vegetables, liver, kidney and tea.

Molybdenum

- Molybdenum is a cofactor for enzymes that convert sulfites to sulfate, and nucleotides to uric acid, and that help metabolize aldehydes & other toxins.
- Low Mo levels may result from long-term TPN that does not include Mo.
- Mo deficiency may result in increased sulfite, decreased plasma uric acid (and antioxidant function), deficient sulfate, impaired sulfation (detoxification), neurologic disorders or brain damage (if severe deficiency).
- Food sources include buckwheat, beans, grains, nuts, beans, lentils, meats and vegetables (although Mo content of plants depends on soil content).

Zinc

- Zinc plays a vital role in immunity, protein metabolism, heme synthesis, growth & development, reproduction, digestion and antioxidant function.
- Low levels may occur with malabsorption, alcoholism, chronic diarrhea, diabetes, excess Cu or Fe, diuretics, ACE inhibitors, H2 blockers or digoxin.
- Deficiency can result in hair loss and skin rashes, also impairments in growth & healing, immunity, sexual function, taste & smell and digestion.
- Food sources include oysters, organ meats, soybean, wheat germ, seeds, nuts, red meat, chicken, herring, milk, yeast, leafy and root vegetables.

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TEST NAME: Metabolomix+

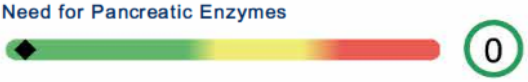
Interpretation At-A-Glance

Microbiome & Digestive Support

Need for Probiotics

- Probiotics have many functions. These include: production of some B vitamins and vitamin K; enhance digestion & absorption; decrease severity of diarrheal illness; modulate of immune function & intestinal permeability.
- Alterations of gastrointestinal microflora may result from C-section delivery, antibiotic use, improved sanitation, decreased consumption of fermented foods and use of certain drugs.
- Some of the diseases associated with microflora imbalances include: IBS, IBD, fibromyalgia, chronic fatigue syndrome, obesity, atopic illness, colic and cancer.
- Food sources rich in probiotics are yogurt, kefir and fermented foods.

Need for Pancreatic Enzymes



- Pancreatic enzymes are secreted by the exocrine glands of the pancreas and include protease/peptidase, lipase and amylase.
- Pancreatic exocrine insufficiency may be primary or secondary in nature. Any indication of insufficiency warrants further evaluation for underlying cause (i.e., celiac disease, small intestine villous atrophy, small bowel bacterial overgrowth).
- A high functional need for digestive enzymes suggests that there is an impairment related to digestive capacity.
- Determining the strength of the pancreatic enzyme support depends on the degree of functional impairment. Supplement potency is based on the lipase units present in both prescriptive and non-prescriptive agents.

Functional Imbalances

Mitochondrial Dysfunction

- Mitochondria are a primary site of generation of reactive oxygen species. Oxidative damage is considered an important factor in decline of physiologic function that occurs with aging and stress.
- Mitochondrial defects have been identified in cardiovascular disease, fatigue syndromes, neurologic disorders such as Parkinson's and Alzheimer's disease, as well as a variety of genetic conditions. Common nutritional deficiencies can impair mitochondrial efficiency.

Need for Methylation

- Methylation is an enzymatic process that is critical for both synthesis and inactivation. DNA, estrogen and neurotransmitter metabolism are all dependent on appropriate methylation activity.
- B vitamins and other nutrients (methionine, magnesium, selenium) functionally support catechol-O-methyltransferase (COMT), the enzyme responsible for methylation.

Toxic Exposure

- Methyl tert-Butyl Ether (MTBE) is a common gasoline additive used to increase octane ratings, and has been found to contaminate ground water supplies where gasoline is stored. Inhalation of MTBE may cause nose and throat irritation, as well as headaches, nausea, dizziness and mental confusion. Animal studies suggest that drinking MTBE may cause gastrointestinal irritation, liver and kidney damage and nervous system effects.
- Styrene is classified by the US EPA as a "potential human carcinogen," and is found widely distributed in commercial products such as rubber, plastic, insulation, fiberglass, pipes, food containers and carpet backing.
- Levels of these toxic substances should be examined within the context of the body's functional capacity for methylation and need for glutathione.

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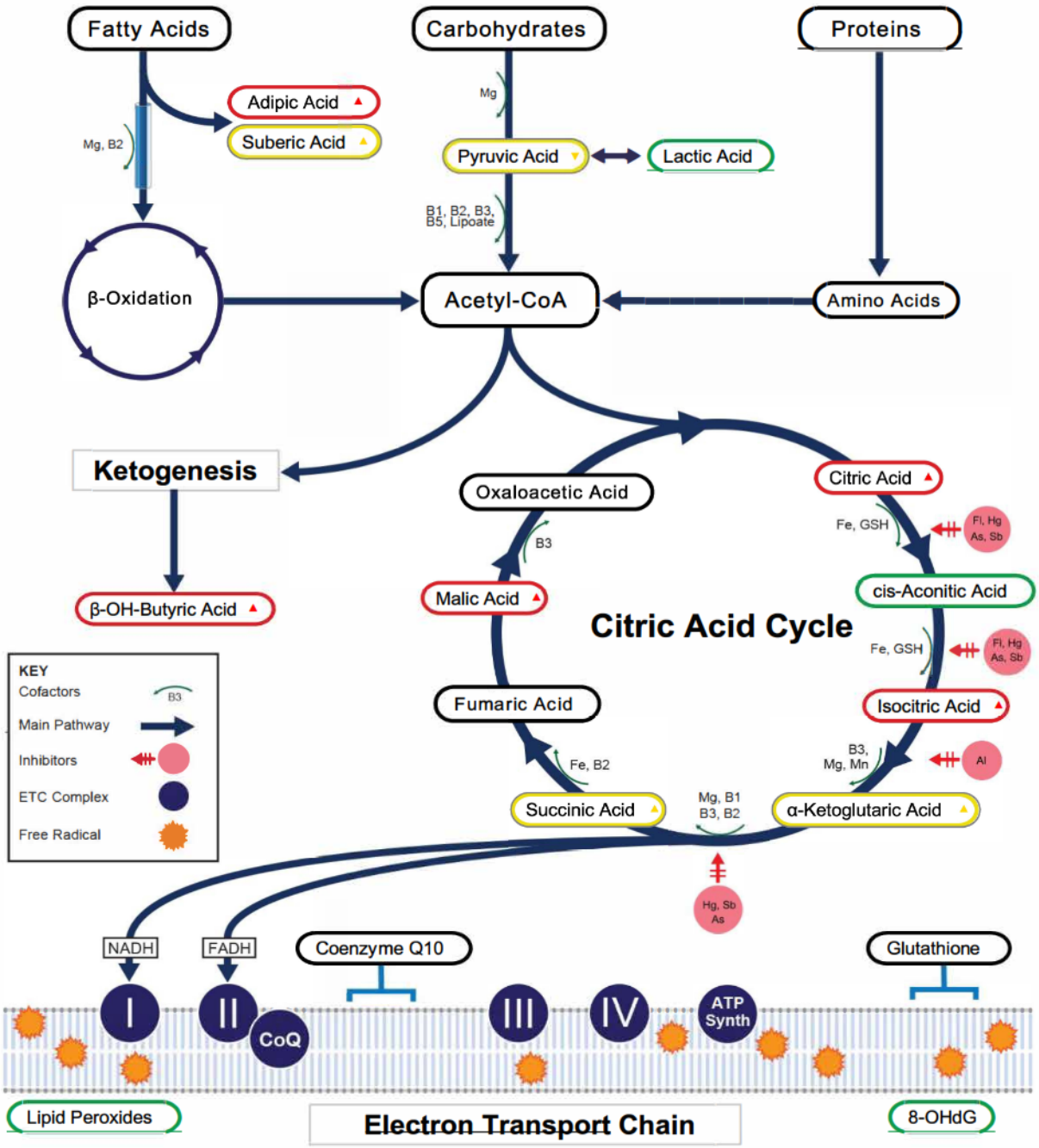
TEST NAME: Metabolomix+

Oxidative Stress & Mitochondrial Dysfunction



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TEST NAME: Metabolomix+

Patient: SAMPLE PATIENT

All biomarkers reported in mmol/mol creatinine unless otherwise noted.

Organic Acids

Malabsorption & Dysbiosis Markers

Malabsorption Markers		Reference Range
Indoleacetic Acid	1.9	<= 4.2
Phenylacetic Acid	0.09	<= 0.12

Dysbiosis Markers		Reference Range
Dihydroxyphenylpropionic Acid (DHPPA)	2.6	<= 5.3
3-Hydroxyphenylacetic Acid	3.1	<= 8.1
4-Hydroxyphenylacetic Acid	7	<= 29
Benzoic Acid	0.04	<= 0.05
Hippuric Acid	<dl	<= 603

Yeast / Fungal Dysbiosis Markers		Reference Range
D-Arabinitol	4	<= 36
Citramalic Acid	15.5	<= 5.8
Tartaric Acid	<dl	<= 15

Vitamin Markers

Branch-Chain Catabolites (B1, B2, B3, ALA)		Reference Range
α-Ketoadipic Acid	0.7	<= 1.7
α-Ketoisovaleric Acid	0.43	<= 0.97
α-Ketoisocaproic Acid	0.47	<= 0.89
α-Keto-β-Methylvaleric Acid	1.8	<= 2.1
Glutaric Acid	0.33	<= 0.51
Isovalerylglycine	2.4	<= 3.7

Methylation Markers (Folate, B12)		
Formiminoglutamic Acid (FIGlu)	1.1	<= 1.5
Methylmalonic Acid	0.8	<= 1.9

Biotin Markers		
3-Hydroxypropionic Acid	14	5-22
3-Hydroxyisovaleric Acid	19	<= 29

Cellular Energy & Mitochondrial Markers

Fatty Acid Metabolism		Reference Range
Adipic Acid	4.2	<= 2.8
Suberic Acid	1.4	<= 2.1

Carbohydrate Metabolism		
Pyruvic Acid	9	7-32
Lactic Acid	9.0	1.9-19.8
α-Hydroxybutyric Acid	11.08	<= 0.83
β-OH-Butyric Acid	>31.0	<= 2.8
β-OH-β-Methylglutaric Acid	12	<= 15

Energy Metabolism		
Citric Acid	998	40-520
cis-Aconitic Acid	23	10-36
Isocitric Acid	93	22-65
α-Ketoglutaric Acid	34	4-52
Succinic Acid	3.6	0.4-4.6
Malic Acid	9.0	<= 3.0

Methodology: GCMS, LC/MS/MS, Alkaline Picrate, Colorimetric

Neurotransmitter Metabolites

Kynurenine Markers (Vitamin B6)		Reference Range
Kynurenic Acid	13.1	<= 7.1
Quinolinic Acid	7.7	<= 9.1
Kynurenic / Quinolinic Ratio	1.70	>= 0.44
Xanthurenic Acid	0.56	<= 0.96

Catecholamine Markers		
Homovanillic Acid	2.8	1.2-5.3
Vanilmandelic Acid	1.8	0.4-3.6
3-Methyl-4-OH-phenylglycol	0.09	0.02-0.22

Serotonin Markers		
5-OH-indoleacetic Acid	9.3	3.8-12.1

Toxin & Detoxification Markers		Reference Range
Pyroglutamic Acid	37	16-34
α-Ketophenylacetic Acid (from Styrene)	0.44	<= 0.46
α-Hydroxyisobutyric Acid (from MTBE)	11.1	<= 6.7
Orotic Acid	0.62	0.33-1.01

Metabolic Analysis Reference Ranges are Age Specific

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TEST NAME: Metabolomix+

Patient: SAMPLE PATIENT

Methodology: Colorimetric, thiobarbituric acid reactive substances (TBARS), Alkaline Picrate, Hexokinase/G-6-PDH, HPLC, GC/MS

Organic Acids

Oxalate Markers		Reference Range
Glyceric Acid	13.8	3.5-16.4
Glycolic Acid	30	<= 67
Oxalic Acid	33	<= 78

All biomarkers reported in mmol/mol creatinine.

Creatinine Concentration		Reference Range
Creatinine ◆	6.2	3.1-19.5 mmol/L

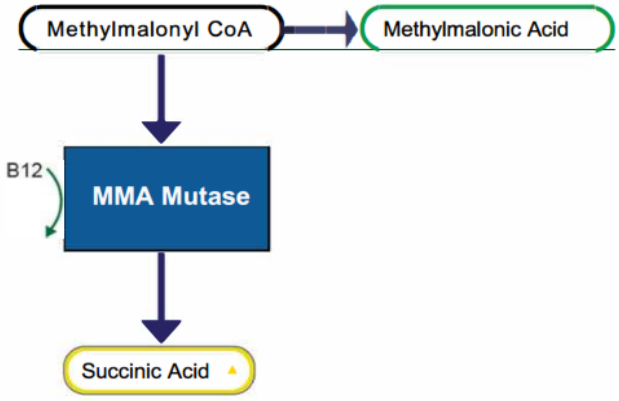
Oxidative Stress Markers

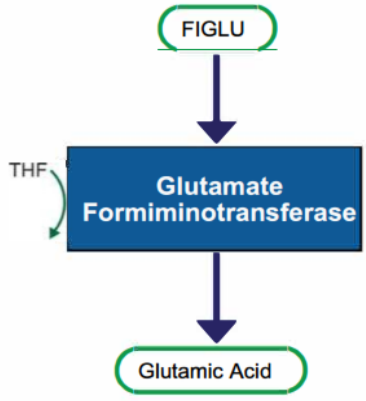
Oxidative Damage		Reference Range
Lipid Peroxides (urine)	5.8	<= 10.0 micromol/g Creat.
8-OHdG (urine)	4	<= 15 mcg/g Creat.

The Oxidative Stress reference ranges are based on an adult population.

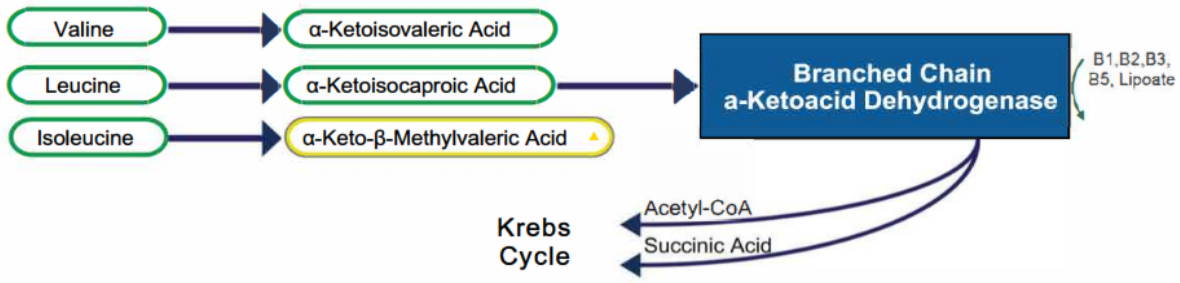
Pathways

Methylation Markers





Branch-Chain Amino Acid Metabolism



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TEST NAME: Metabolomix+

All biomarkers reported in micromol/g creatinine unless otherwise noted.

Amino Acids (FMV)

Nutritionally Essential Amino Acids

Amino Acid		Reference Range
Arginine	<dl	3-43
Histidine	361	124-894
Isoleucine	11	3-28
Leucine	22	4-46
Lysine	36	11-175
Methionine	5	2-18
Phenylalanine	25	8-71
Taurine	266	21-424
Threonine	78	17-135
Tryptophan	23	5-53
Valine	32	7-49

Nonessential Protein Amino Acids

Amino Acid		Reference Range
Alanine	41	63-356
Asparagine	88	25-166
Aspartic Acid	<dl	<= 14
Cysteine	26	8-74
Cystine	25	10-104
γ-Aminobutyric Acid	1	<= 5
Glutamic Acid	15	4-27
Glutamine	188	110-632
Proline	3	1-13
Tyrosine	44	11-135

Creatinine Concentration

		Reference Range
Creatinine◆	6.1	3.1-19.5 mmol/L

Intermediary Metabolites

B-Vitamin Markers		Reference Range
α-Aminoadipic Acid	16	2-47
α-Amino-N-butyric Acid	14	2-25
β-Aminoisobutyric Acid	149	11-160
Cystathionine	3	2-68

Urea Cycle Markers		
Citrulline	2.3	0.6-3.9
Ornithine	8	2-21
Urea◆	227	168-465 mmol/g creatinine

Glycine/Serine Metabolites		
Glycine	250	95-683
Serine	132	40-163
Ethanolamine	190	50-235
Phosphoethanolamine	4	1-13
Phosphoserine	<dl	3-13
Sarcosine	1.1	<= 1.1

Dietary Peptide Related Markers		Reference Range
Anserine (dipeptide)	0.4	0.4-105.1
Carnosine (dipeptide)	3	1-28
1-Methylhistidine	12	38-988
3-Methylhistidine	38	44-281
β-Alanine	4	<= 22

Amino Acid reference ranges are age specific.

Methodology: LC/MS/MS, Alkaline Picrate

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